Core Mathematics

A Practical Guide on Mathematics for Physics Learners

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These are lecture notes for first-year university students in physics or physical science. They cover the mathematical tools you need most: calculus, logic, complex numbers, and vectors, with introductions to vector calculus, differential equations, and statistical analysis. Exercises are central to these notes. Practice each topic repeatedly until you can work through problems quickly and accurately.

Preface

Physics is an activity to describe the world, but how? At present, mathematics (+ *broken* English) is the only language we can use. So, unfortunately, we need to learn mathematics! (and English...🥺) But, in some sense, you are fortunate, since you have a language. Math and physics have developed together, so you will understand physics better if you study mathematics well!

When you do DIY, you need to use the tools properly, smoothly, and accurately. It is the same here; when we use math as a tool, it must be **logical**, **quick**, and **accurate**. To be logical, you must think each problem carefully; otherwise your discussion might be incomplete and you can't convince others. To be quick and accurate, you must *"drill"* repeatedly, as if top NBA players do shooting practice almost everyday. You should reach the point where basic calculations feel automatic.

This is not a math textbook, so theorem proofs are mostly omitted. **This is rather** an exercise book. Problems and drills are the core of this book.

Target of this document

This document is primarily for first-year undergraduate students in their second semester, who are majoring in physics or physical science.

How to use this book

First, buy a A4-sized paper notebook. You are expected to solve problems, and when you solve "problems", you are strongly recommended to write down your solution process on the notebook, preferably with full descriptions in English. Meanwhile, "drills" are for you to practice calculations, so you can just write down the final answer, or just do it in your brain.

Prepare the textbook [Boas] (see "References" below) and use it as a reference. When you want to understand the details of a topic, or when you want to solve more problems, consult [Boas].

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Visit https://github.com/misho104/LecturePublic for further information, updates, and to report issues.

References

This document is written under the influence of the following references.

- The Math Book by Hal Tasaki (数学：物理を学び楽しむために, in Japanese)

  A free book with full description, but unfortunately in Japanese. Topic selections and the depth of descriptions are based on this reference.

- First-Year Math Practice (in Japanese)

  This document is grounded in a "learn steadily" approach ("じっくり" in Japanese), which originates from Prof. Gocho and Prof. Kiyono, who were instructors of Sho in his freshman. Problems and rigorous math descriptions are taken from their works.

- Mary L. Boas, *Mathematical Methods in the Physical Sciences*, 3rd ed., Wiley, 2006.
- Arfken, Weber, and Harris, *Mathematical Methods for Physicists*, 7th ed., Academic Press, 2023.

  These two books are widely used in physics departments around the world and referred to in this document many times as [Boas] and [AWH]. [Boas] seems more friendly to beginners, while [AWH] seems more detailed and complete. Sho recommends you to study [Boas] *in parallel with* this document, using it as a reference, and to consult [AWH] when you want more details or more exercises.

Lecture Plan

The content is designed for 150-minute × 14-week lectures, as it is originally for a lecture *Mathematics for Fundamental Physics* (基礎物理數學) in National Sun Yat-sen University (國立中山大學).

1. Derivative
2. Units. Significant Digits.
3. Logic. Theorems. a^x and $\log_a x$.
4. Logic. Theorems. a^x and $\log_a x$.
5. Vectors are arrows.
6. $a \cdot b$ and $a \times b$
7. Matrices (rotation, reflection, scaling)
8. Complex numbers (polar form, Euler's formula, multi-valuedness)
9. Complex vectors, Hermitian inner product, bra-ket notation
10. Vector calculus (grad, div, rot; example of point charge)
11. ODE basics
12. ODE basics
13. Probability and error analysis
14. Probability and error analysis

Symbols

Mathematical Notations

Chapter 1	
$\mathbb{N}$	natural numbers
$\mathbb{N}_0$	natural numbers (0, 1, 2, ...)
$\mathbb{N}^+$	natural numbers (1, 2, ...)
$\mathbb{R}$	real numbers
$\mathbb{Q}$	rational numbers
$\mathbb{C}$	complex numbers

Greek symbols

A	α	alpha
B	β	beta
Γ	γ	gamma
Δ	δ	delta
E	ε	epsilon
Z	ζ	zeta

H	η	eta
Θ	θ	theta
I	ι	iota
K	κ	kappa
Λ	λ	lambda
M	μ	mu

N	ν	nu
Ξ	ξ	xi
O	o	omicron
Π	π	pi
P	ρ	rho
Σ	σ	sigma

T	τ	tau
Y	υ	upsilon
Φ	$\varphi\,(\phi)$	phi
X	χ	chi
Ψ	ψ	psi
Ω	ω	omega

Gray-outed ones are almost never used, but others are almost always used. You need to write them so that **others can distinguish each from others**, but Sho thinks Japanese- or Chinese-speakers are OK with it, as we everyday read and write 日 vs 日 vs 白 or 土 vs 士 vs 工 properly!

Quiz

★**1.** Read out the following words.

(1) α-particle **(2)** β-decay **(3)** Γ-function
(4) ε-δ definition **(5)** angle θ and φ **(6)** μ- and τ-leptons
(7) $\psi(x)$ and $\varphi(x)$ **(8)** ζ and ξ **(9)** metric $\eta_{\mu\nu}$
(10) X-ray and γ-ray **(11)** Ω-baryon **(12)** $\varepsilon_{\mu\nu\rho\sigma}$-tensor

Problems

★**0.1.** Write the following letters so that people can distinguish each from others.

(1) a, α	**(2)** b, β	**(3)** c, C	**(4)** e, E, ε
(5) ζ, ξ	**(6)** $\Theta, \theta, Q, \sigma, 6$	**(7)** $g, q, 9$	**(8)** $i, l, 1, I$
(9) k, K, κ	**(10)** m, μ	**(11)** p, ρ	**(12)** s, S
(13) t, τ, T	**(14)** u, v, ν	**(15)** w, W, ω	**(16)** x, X, χ

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tangent line 9
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Chapter 1

Derivative (Review)

1.1 Introduction

Remark: This document is designed for first-year students in their *second semester*, who **have already studied derivatives** in their first-semester *"calculus"* course. If you are not yet confident with derivatives, please study the Derivative Boot Camp first[#1].

Birds sing, fish swim, flowers bloom, stars twinkle, and university students calculate derivatives. Let us begin with basic **derivatives**. Try the next quiz—and note how long it takes.

Quiz

★**1.** Calculate the first derivatives of the following functions.

(1) $(x+1)^3$ **(2)** $\tan x$ **(3)** $2\cos^2 x$ **(4)** $2\cos x^2$

(5) $x^3 \cos x$ **(6)** $\dfrac{3\sin x}{x}$ **(7)** $x^{3/2}$ **(8)** $\sqrt{\sin x}$

If you finished within two minutes—Wonderful! Most students take 3–4 minutes. If you took more than four minutes, do not worry: you can do the calculations, and a little more practice will help. Try some problems in the Derivative Boot Camp

Did you make calculation mistakes and get wrong answers? That is a serious issue—just as serious as forgetting how to do the calculations at all. You will perform similar calculations more than 100 times in university, so you need to be both fast and accurate.

At university, students often underestimate the importance of basic calculations. In physics, simple calculations appear constantly, so your speed and accuracy directly affect how well you follow lectures, how efficiently you study, and ultimately your grade. Both can be improved with practice.

This course will help you strengthen these basic calculations—some of which you already know from high school—while also introducing new topics that are essential for physics. You will solve many problems and drills, just as a beginner athlete repeats the same move hundreds of times to master it.

[#1] Visit https://misho104.github.io/LecturePublic and find `gp1_boot1_deriv_true.pdf`.

1.2 Typical Pitfalls

1.2.1 Notations

There are several equivalent ways to write the derivative of $f(x)$:

$$f'(x) = \frac{\mathrm{d}f}{\mathrm{d}x}(x) = \frac{\mathrm{d}f(x)}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}x}f(x) \qquad \text{[All means the same thing].} \tag{1.1}$$

Similarly, the value of $f'(x)$ at a specific point $x = 3$ can be written as

$$f'(3) = f'(x)\Big|_{x=3} = \frac{\mathrm{d}f}{\mathrm{d}x}\Big|_{x=3} = \frac{\mathrm{d}f}{\mathrm{d}x}(3) = \frac{\mathrm{d}}{\mathrm{d}x}f(3), \tag{1.2}$$

and they are all equivalent. For example, if $f(x) = 3x^2 + 6$, then $f'(x) = 6x$ and $f'(3) = 18$.

The real pitfalls come next. Students often get confused when a **constant** a appears. If a is *declared as a constant*, we can define a function such as $g(x) = ax^2 + 2a$, for which

$$g'(x) = 2ax. \tag{1.3}$$

We can evaluate $g'(x)$ at $x = a$. The result is $g'(a) = 2a^2$, written as

$$g'(a) = g'(x)\Big|_{x=a} = \frac{\mathrm{d}g}{\mathrm{d}x}(a) = \frac{\mathrm{d}}{\mathrm{d}x}g(a) = 2a^2. \tag{1.4}$$

Be careful: the notation $\frac{\mathrm{d}}{\mathrm{d}x}g(a)$ does **not** mean $\frac{\mathrm{d}}{\mathrm{d}x}\big[g(a)\big] = \frac{\mathrm{d}}{\mathrm{d}x}\big(a^3 + 2a\big) = 0$.

Fail safe note: If you can't see Eq. (1.3), try setting $a = 3$. Then, $g(x) = f(x)$, and $g'(x)$ should equal $f'(x)$.

There is more bad news. *Physicists are often lazy*, and we frequently omit the argument (x). For example, if $f(x) = 3x^2$ and $F(t) = t^2 + at + 1$, and if a is *declared as a constant*,

$$\begin{aligned} f' &= \frac{\mathrm{d}}{\mathrm{d}x}f = 6x, & f'(a) &= \frac{\mathrm{d}}{\mathrm{d}x}f(a) = 6a, \\ F' &= \frac{\mathrm{d}}{\mathrm{d}t}F = 2t + a, & F'(a) &= \frac{\mathrm{d}}{\mathrm{d}t}F(a) = 3a. \end{aligned} \tag{1.5}$$

So when you see a function written as f, you must identify its variable by reading the context. For example, we sometimes replace the variables. Assume $f(x) = 3x^2$. If t is another **variable** (= not a constant), we can understand $3t^2$ is the same function $f(t)$. So we may write $f(t) = 3t^2$, whose derivative is $f'(t) = \mathrm{d}f/\mathrm{d}t = 6t$.

Advanced note: We usually write "Let k be a constant" to declare k as a constant. You need to do so, too.

Higher-order derivatives is written as

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}f}{\mathrm{d}x}\right) = \frac{\mathrm{d}^2 f}{\mathrm{d}x^2} = f''(x) = f^{(2)}(x), \qquad \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}f}{\mathrm{d}x}\right)\right) = \frac{\mathrm{d}^3 f}{\mathrm{d}x^3} = f'''(x) = f^{(3)}(x), \tag{1.6}$$

etc. Please be careful on the position of "2" and "3".

Quiz

★**2.** Let a and b be constants. If $f(x) = 3x^2$, $g(t) = at^2 + b$, $F(x) = a + 3$, and $G(t) = a\cos(t) + x\sin(t)$, what are the following expressions?

(1) $f'(x)$	**(2)** $f'(0)$	**(3)** $f'(a)$	**(4)** $f'(-a)$
(5) $g'(t)$	**(6)** $g'(0)$	**(7)** $g'(b)$	**(8)** $g'(x)$
(9) $F'(a)$	**(10)** $G'(t)$	**(11)** $G'(0)$	**(12)** $G'(a)$
(13) $\frac{\mathrm{d}f}{\mathrm{d}x}$	**(14)** $\frac{\mathrm{d}f}{\mathrm{d}x}(0)$	**(15)** $\frac{\mathrm{d}g}{\mathrm{d}t}$	**(16)** $\frac{\mathrm{d}g(x)}{\mathrm{d}x}$

1.2.2 Radian and trigonometric functions

At university, angles are almost always measured in **radians**:

$$360 \text{ degree} \quad (\text{or: } 360^\circ) \quad = \quad 2\pi \text{ radian} \quad (\text{or: } 2\pi \text{ rad}) \tag{1.7}$$

Furthermore, we usually omit "radian" (because we are lazy!) Now,

$$\text{a right angle is } \pi/2. \quad \text{The sum of the interior angles of a triangle is } \pi. \tag{1.8}$$

Quiz

★**3.** Express the following angles in radians, and radians in angles.

(1) 180°	**(2)** 45°	**(3)** 60°	**(4)** 90	**(5)** -30°
(6) 1°	**(7)** 1	**(8)** -0.3	**(9)** x°	**(10)** x

Why do insist on radians? The answer comes from the derivative formula

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin x = \cos x. \tag{1.9}$$

because we choose radians as the standard, $(\sin x)'$ becomes this simple.

Advanced note: The simpleness of Eq. (1.9) originates in the fact $\sin(0.01 \text{ rad}) \approx 0.01$. If we used degrees, we would have $\sin(0.01^\circ) \approx 0.01 \times 0.017453$ and everything is messed up with this number 0.017453.

There are a few remarks in the notation of **trigonometric functions**:

$$\begin{aligned} &\sin^2 x \neq \sin x^2. \quad \text{Namely,} \quad (\sin x)^2 = \sin^2 x \quad \neq \quad \sin x^2 = \sin(x^2). \\ &\tan^{-1} x \neq \frac{1}{\tan x}. \quad \text{Namely,} \quad \tan^{-1} x = \arctan x \quad \neq \quad (\tan x)^{-1} = \frac{1}{\tan x} = \cot x. \end{aligned} \tag{1.10}$$

The following are not incorrect but confusing. *We should avoid ambiguity*, so please don't use them.

$$\sin^{-2} x, \quad \sin^{1/2} x, \quad \sin(x)^2, \quad \sin(x+1)^2, \ldots$$

Sho thinks we should only use $\sin^k x$ for $k = 2, 3, 4, \ldots$, and write $\arcsin x$ instead of $\sin^{-1} x$.

1.2.3 Several interpretations of derivatives

Consider a function $f(t)$. When you discuss its derivative $f'(t)$, you physicists should have the following three interpretations:

- Regarding t as the time, $f'(t)$ is the **rate of change** of $f(t)$ per unit time. If $f'(t) > 0$, the function f is increasing at the time t, while $f'(t) < 0$ means it is decreasing.
- Consider a graph of $f(t)$. Then, $f'(t)$ is the slope of the **tangent line** to the graph at the point t.
- Mathematically, $f'(t)$ is defined by (Check that they are equivalent.)

$$f'(t) := \lim_{\Delta t \to 0} \frac{f(t+\Delta t) - f(t)}{\Delta t} = \lim_{h \to 0} \frac{f(t+h) - f(t-h)}{2h} = \lim_{s \to t} \frac{f(s) - f(t)}{s - t}. \tag{1.11}$$

For further description, please consult other textbooks or online resources.

1.3 Taylor expansion

Reviewing the previous equation for a function $f(x)$, we have

$$f'(x) := \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}, \qquad f'(a) := \lim_{\delta \to 0} \frac{f(a+\delta) - f(a)}{\delta}. \tag{1.12}$$

Advanced note: These two have a bit different sense: the first one defines a new function $f'(x)$, while the second one defines a number that is eventually equal to $f'(x)|_{x=a}$. Anyway, we don't care the difference.

We can understand the second equation as

$$\text{If } \delta \approx 0, \quad f'(a) \approx \frac{f(a+\delta) - f(a)}{\delta}, \quad \text{i.e.,} \quad f(a+\delta) \approx f(a) + \delta f'(a). \tag{1.13}$$

These equations will be helpful if you want to know **what is $f(x)$ around the point $x = a$?**

Example 1.1

Find the approximate value of the following expressions:

(1) $\sqrt{1.002}$ **(2)** $(1.002)^{10}$ **(3)** $\sqrt{4.004}$

Solution

(1) Apply Eq. (1.13) for $f(x) = \sqrt{x}$, $a = 1$, and $\delta = 0.002$. Then,

$$f'(x) = \frac{1}{2\sqrt{x}}, \quad f'(a) = f'(1) = \frac{1}{2}, \quad f'(a) \approx f(a) + \delta f'(a) = \sqrt{1} + \frac{\delta}{2} = 1.001.$$

(2) Doing the same thing for $g(x) = x^{10}$, we have

$$g'(x) = 10x^9, \quad g'(a) = g'(1) = 10, \quad g'(a) \approx g(a) + \delta g'(a) = 1^{10} + 10\delta = 1.02.$$

(3) Doing the same thing for $h(x) = \sqrt{x}$, $a = 4$, and $\delta = 0.004$, we have

$$h'(4) = \frac{1}{4}, \quad h'(4.004) \approx h(4) + 0.004 \times \frac{1}{4} = 2.001.$$

Theorem 1.2 (Taylor's theorem)

If a function $f(x)$ is sufficiently smooth, we can express it as

$$f(x) = \sum_{k=0}^{N} \frac{(x-a)^k}{k!} f^{(k)}(a) + [\text{dangerous part}]. \tag{1.14}$$

and we can ignore the dangerous part if $x \approx a$. So, if we replace x by $a + \varepsilon$ and $\varepsilon \approx 0$,

$$\begin{aligned} f(a+\varepsilon) &= \sum_{k=0}^{N} \frac{\varepsilon^k}{k!} f^{(k)}(a) + [\text{dangerous part}] \\ &\approx f(a) + \varepsilon f'(a) + \frac{\varepsilon^2}{2} f''(a) + \frac{\varepsilon^3}{6} f'''(a) + \cdots \end{aligned} \tag{1.15}$$